Spectrum
SPC

From Binary to Beyond

بنك أسئلة بايثون

50 سؤال اختيار من متعدد · Python Fundamentals

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الأساسيات

1 What is the type of the result of 10 / 2 in Python 3?

A int
B float
C str
D bool

2 What is the output of the following code?

```
x = 5
x = "five"
x = 2.5
print(type(x))
```

A <class 'int'>
B <class 'str'>
C <class 'float'>
D Error, because the type of x changed

3 What is the output of the following code?

```
x = 5
x += 3
x *= 2
print(x)
```

A 16
B 13
C 11
D 10

4 What is the output of the following code?

```
print(2 + 3 * 4 ** 2 / 8)
```

A 10.0
B 8
C 2.5
D 8.0

5 What is the output of the following code?

```
print(not (5 > 3 and 2 > 4) or False)
```

A False
B True
C Error
D None

6 What is the output of the following code?

```
print(int(3.9) + int("2"))
```

A 5.9
B 6
C 5
D Error

7 What is the output of the following code?

```
print(bool(0), bool("0"), bool(""), bool([]))
```

A False True False False
B False False False False
C False True True False
D True True False False

8 What is the output of the following code?

```
print(7 // 2, 7 % 2, 7 ** 2)
```

A 3.5 1 14
B 3 1 14
C 4 1 49
D 3 1 49

الشروط

9 What is printed by the following code?

```
x = 15
if x > 10:
    print("A")
if x > 12:
    print("B")
elif x > 5:
    print("C")
else:
    print("D")
```

A A only
B A and C on separate lines
C A and B on separate lines
D A, B and C on separate lines

10 Which condition is True only when n is even AND between 10 and 50 (inclusive)?

A n % 2 == 0 or 10 <= n <= 50
B n % 2 == 0 and 10 <= n <= 50
C n % 2 == 1 and 10 < n < 50
D n / 2 == 0 and 10 <= n <= 50

11 What is printed by the following code?

```
age = 20
has_id = False
if age >= 18:
    if has_id:
         print("Enter")
    else:
         print("Show ID")
else:
    print("Too young")
```

A Show ID
B Enter
C Too young
D Nothing is printed

الحلقات

12 What is the output of the following code?

```
for i in range(2, 10, 3):
    print(i, end=" ")
```

A 2 5 8 11
B 2 5 8 10
C 2 3 4 5 6 7 8 9
D 2 5 8

13 What is the value of t after the following code runs?

```
t = 0
for i in range(5, 0, -1):
    t += i
print(t)
```

A 10
B 15
C 14
D 0

14 What is the output of the following code?

```
n = 1
while True:
    n *= 2
    if n > 50:
        break
print(n)
```

A 32
B 50
C 64
D 128

15 What is the output of the following code?

```
for i in range(1, 8):
    if i % 3 == 0:
        continue
    if i == 7:
        break
    print(i, end=" ")
```

A 1 2 4 5
B 1 2 4 5 7
C 1 2 3 4 5 6
D 1 2 4 5 6

16 What does the following code print?

```
for i in range(1, 4):
    for j in range(i):
        print("*", end="")
    print()
```

A Three lines, each containing ***
B One line: ******
C *** then ** then * (three lines)
D * then ** then *** (three lines)

17 What is the output of the following code?

```
count = 0
for i in range(3):
    for j in range(4):
        if j == 2:
            break
        count += 1
print(count)
```

A 12
B 6
C 9
D 8

18 What is the output of the following code?

```
t = 0
for n in range(1, 11):
    if n % 2 == 0:
        t += n
print(t)
```

A 25
B 55
C 30
D 20

19 What is the output of the following code?

```
i = 10
while i > 0:
    i -= 3
print(i)
```

A -2
B 1
C 0
D -3

القوائم

20 What is the output of the following code?

```
print([1, 2] + [3] * 2)
```

A [1, 2, 6]
B [1, 2, 3]
C [1, 2, [3, 3]]
D [1, 2, 3, 3]

21 What is the output of the following code?

```
a = [3, 1, 2]
b = a.sort()
print(b)
```

A [1, 2, 3]
B None
C [3, 1, 2]
D Error

22 What is the output of the following code?

```
a = [3, 1, 2]
b = sorted(a, reverse=True)
print(a, b)
```

A [1, 2, 3] [3, 2, 1]
B [3, 2, 1] [3, 2, 1]
C [3, 1, 2] [3, 2, 1]
D [3, 1, 2] [3, 1, 2]

23 What is the final content of lst?

```
lst = [3, 1, 2]
lst.append(5)
lst.insert(1, 9)
lst.remove(1)
print(lst)
```

A [3, 9, 2, 5]
B [3, 9, 1, 2, 5]
C [9, 3, 2, 5]
D [3, 1, 2, 5, 9]

24 What is the output of the following code?

```
lst = [4, 8, 15, 16]
x = lst.pop()
y = lst.pop(0)
print(x, y, lst)
```

A 16 8 [4, 15]
B 4 16 [8, 15]
C 15 4 [8, 16]
D 16 4 [8, 15]

25 What is the output of the following code?

```
nums = [5, 10, 15, 20, 25]
print(nums[1:4], nums[::2])
```

A [10, 15, 20, 25] [5, 15, 25]
B [10, 15, 20] [5, 15, 25]
C [10, 15, 20] [5, 10, 15]
D [5, 10, 15, 20] [10, 20]

26 What is the output of the following code?

```
d = [4, 9, 2, 7]
print(len(d), sum(d), max(d) - min(d))
```

A 4 20 7
B 4 22 5
C 4 22 7
D 3 22 7

27 What is the output of the following code?

```
nums = [4, 17, 9, 17, 3]
m = nums[0]
for n in nums:
    if n > m:
        m = n
print(m, nums.index(m))
```

A 17 1
B 17 3
C 17 4
D 9 2

28 What is the output of the following code?

```
print([x ** 2 for x in range(1, 6) if x % 2 == 1])
```

A [1, 4, 9, 16, 25]
B [4, 16]
C [1, 3, 5]
D [1, 9, 25]

29 What is the output of the following code?

```
matrix = [[1, 2, 3],
          [4, 5, 6],
          [7, 8, 9]]
print(matrix[1][2], matrix[2][0])
```

A 8 3
B 6 7
C 7 6
D 5 9

30 What is the output of the following code?

```
matrix = [[1, 2, 3],
          [4, 5, 6],
          [7, 8, 9]]
t = 0
for i in range(3):
    t += matrix[i][i]
print(t)
```

A 6
B 18
C 15
D 12

31 What is the output of the following code?

```
a = [1, 2, 3]
b = a
c = a.copy()
b.append(4)
c.append(5)
print(a, c)
```

A [1, 2, 3, 4] [1, 2, 3, 5]
B [1, 2, 3] [1, 2, 3, 5]
C [1, 2, 3, 4, 5] [1, 2, 3, 5]
D [1, 2, 3, 4] [1, 2, 3, 4, 5]

32 What is the output of the following code?

```
result = []
for n in range(1, 8):
    if n % 2 == 0:
        result.append(n * n)
print(result)
```

A [2, 4, 6]
B [4, 16, 36, 64]
C [1, 9, 25, 49]
D [4, 16, 36]

33 What is the output of the following code?

```
words = ["a", "bb", "ccc", "dd"]
count = 0
for w in words:
    if len(w) == 2:
        count += 1
print(count)
```

A 1
B 2
C 3
D 4

34 What is the output of the following code?

```
nums = [1, 2, 3, 4, 5]
print(nums[::-1][:2])
```

A [1, 2]
B [4, 5]
C [5, 4]
D [2, 1]

35 What is the output of the following code?

```
a = [5, 3, 8, 1]
for i in range(len(a) - 1):
    if a[i] > a[i + 1]:
         a[i], a[i + 1] = a[i + 1], a[i]
print(a)
```

A [3, 5, 1, 8]
B [1, 3, 5, 8]
C [3, 5, 8, 1]
D [5, 3, 8, 1]

36 What is the output of the following code?

```
data = [1, 2, 2, 3, 3, 3]
unique = []
for d in data:
    if d not in unique:
        unique.append(d)
print(unique)
```

A [1, 2, 2, 3, 3, 3]
B [1, 2, 3, 3]
C [3, 3, 3]
D [1, 2, 3]

الدوال

37 What is the output of the following code?

```
def f(x):
    print(x * 2)
r = f(4)
print(r)
```

A 8 then 8
B 8 then None
C None then 8
D 4 then None

38 What is the output of the following code?

```
def calc(a, b=3):
    return a * b
print(calc(4), calc(4, 2))
```

A 12 12

B 8 8

C 12 8

D Error, because b is missing in the first call

39 What is the output of the following code?

```
def min_max(lst):
    return min(lst), max(lst)
lo, hi = min_max([5, 2, 9])
print(lo + hi)
```

A 11

B 7

C 14

D (2, 9)

40 What is the output of the following code?

```
def count_even(nums):
    c = 0
    for n in nums:
        if n % 2 == 0:
            c += 1
    return c
print(count_even([1, 2, 4, 7, 10]))
```

A 2

B 5

C 4

D 3

41 What is the output of the following code?

```
x = 5
def change():
    x = 10
    return x
change()
print(x)
```

A 10
B 5
C Error
D None

42 What is the output of the following code?

```
def check(n):
    for i in range(2, n):
        if n % i == 0:
            return False
    return True
print(check(7), check(9))
```

A True True
B False True
C True False
D False False

43 What does the following code print?

```
def digit_sum(n):
    t = 0
    while n > 0:
        t += n % 10
        n //= 10
    return t
print(digit_sum(472))
```

A 13
B 4
C 472
D 11

44 What is the output of the following code?

```
def add_item(lst):
    lst.append(99)
nums = [1]
add_item(nums)
print(nums)
```

A [1]
B [99]
C Error
D [1, 99]

45 What is the output of the following code?

```
def first_neg(lst):
    for x in lst:
        if x < 0:
            return x
    return None
print(first_neg([3, 5, -2, -8]))
```

A -8
B -2
C None
D 3

46 What is the output of the following code?

```
def sq(x):
    return x * x
def sum_sq(a, b):
    return sq(a) + sq(b)
print(sum_sq(3, 4))
```

A 7
B 12
C 25
D 49

47 What is the output of the following code?

```
def fact(n):
    r = 1
    while n > 1:
        r *= n
        n -= 1
    return r
print(fact(5))
```

A 120
B 24
C 5
D 720

مفاهيم

48 Which statement about the return keyword is correct?

A return prints the value on the screen
B A function can contain only one return statement
C Statements after return on the same path still run
D return ends the function and sends a value back to the caller

49 Which loop is the most suitable for repeating a task until the user enters the correct password, when the number of attempts is unknown?

A for loop with range()
B while loop
C Nested for loop
D None of them can be used

50 What is the output of the following code?

```
def above_avg(nums):
    avg = sum(nums) / len(nums)
    return [n for n in nums if n > avg]
print(above_avg([10, 20, 30, 40]))
```

A [10, 20]

B [25]

C [30, 40]

D [20, 30, 40]

مفتاح الإجابات

1	B	2	C	3	A	4	D	5	B
6	C	7	A	8	D	9	C	10	B
11	A	12	D	13	B	14	C	15	A
16	D	17	B	18	C	19	A	20	D
21	B	22	C	23	A	24	D	25	B
26	C	27	A	28	D	29	B	30	C
31	A	32	D	33	B	34	C	35	A
36	D	37	B	38	C	39	A	40	D
41	B	42	C	43	A	44	D	45	B
46	C	47	A	48	D	49	B	50	C